



ASSIGNMENT

Module: 02

Software Development Life Cycle (SDLC) in Age of AI

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Module-02

Software Development Life Cycles (SDLC) in the Age of AI

Q1. Draw the SDLC lifecycle diagram and briefly explain the workflow.

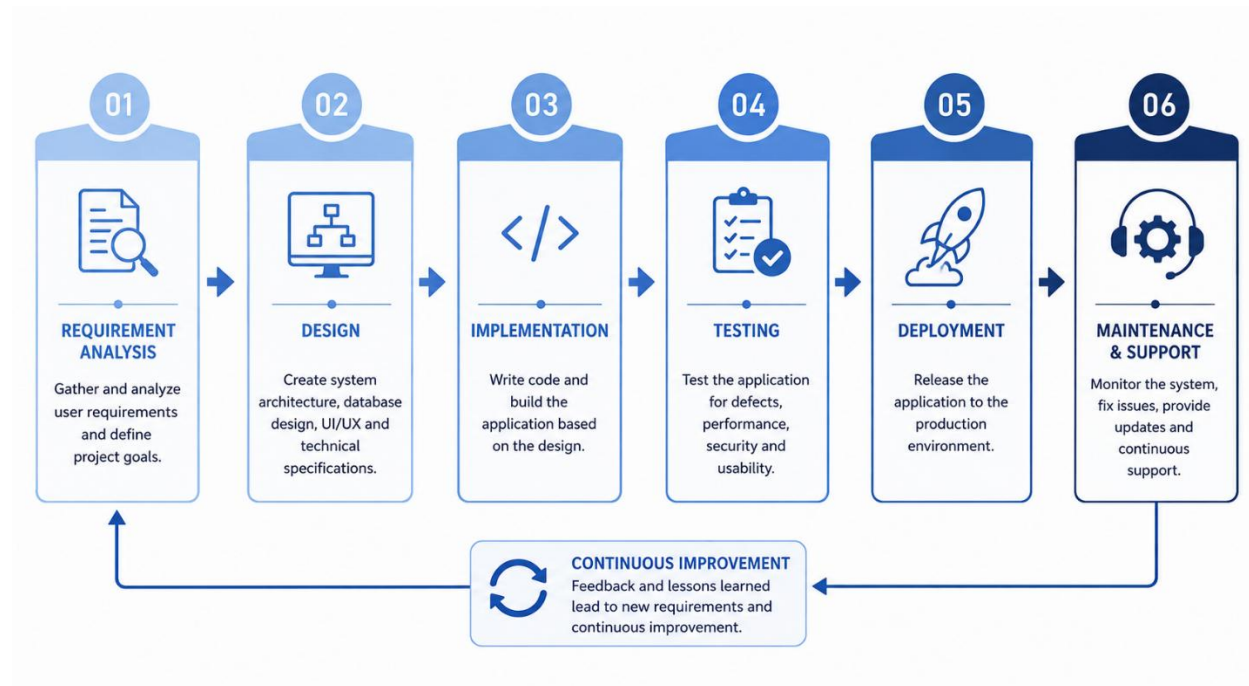


Figure 01: SDLC Life Cycle Diagram

SDLC Workflow

The Software Development Life cycle is a systematic process used to develop, test, deploy, and maintain software. The workflow moves through the following stages:

1. Requirement Analysis:

The development team identifies and analyzes the needs of users and stakeholders. Requirements are collected, documented, prioritized and validated to clearly define what the software should accomplish.

2. Design:

The requirements are converted into a technical design. This phase includes designing the software architecture, database, user interface, system components, and security structure.

3. **Implementation:**

Developers convert the approved design into working software by writing and integrating the source code. Code reviews and development tools, including AI coding assistants, may be used to improve productivity.

4. **Testing:**

The software is systematically tested to identify defects and verify that it meets the specified requirements. Testing may include functional, integration, regression, performance, usability, and security testing.

5. **Deployment:**

After successful testing and approval, the software is released to the production environment where users can access and use the application.

6. **Maintenance & Support:**

After deployment, the software is continuously monitored and maintained. The team fixes bugs, addresses security issues, improves performance, and introduces new features based on user feedback.

Continues Feedback:

SDLC is not always strictly linear, especially in Agile development. Feedback from users, testing, and production can lead to new or changed requirements. These requirements start another development cycle.

Q2. Practical AI Exercise

Selected AI Tool: ChatGPT. The following exercise demonstrates how AI can support requirements engineering and Scrum planning for an E-commerce application.

2.1 Five User Stories for an E-commerce Application

User stories follow the standard format: “As a [user], I want [goal], so that [benefit].”

ID	Features	User Story
User 01	User Registration	As a customer, I want to create an account using my email address and password, so that I can securely access the e-commerce application and manage my orders.
User 02	Product Search	As a customer, I want to search for products by name or keyword, so that I can quickly find the products I am looking for.
User 03	Add Product to Cart	As a customer, I want to add products to my shopping cart, so that I can purchase multiple products in a single order.
User 04	Place an Order	As a customer, I want to place an order by providing my delivery and payment information, so that I can purchase the selected products.
User 05	Track an Order	As a customer, I want to track the status of my order, so that I know when my order will be delivered.

2.2 Acceptance Criteria for US-03: Add Product to Cart

Selected user story: “As a customer, I want to add products to my shopping cart, so that I can purchase multiple products in a single order.

ID	Acceptance Criterion
AC 01	Product can be added: The customer should be able to add an available product to the shopping cart by clicking the “ Add to Cart ” button
AC 02	Cart updates: After adding a product, the cart should display the product name, price, quantity, and subtotal .
AC 03	Quantity can be changed: The customer should be able to increase or decrease the product quantity in the cart.
AC 04	Stock limit is enforced: The system should not allow the customer to add a quantity greater than the available stock .
AC 05	Total price is calculated: The cart should automatically update the total price whenever the product quantity changes
AC 06	Product can be removed: The customer should be able to remove a product from the shopping cart.
AC 07	Empty cart is handled: If all products are removed, the system should display an “ Your cart is empty ” message.

2.3 AI-Generated Sprint Backlog

The following backlog is designed for a two-week sprint. Priority, story points, and tasks are indicative and should be refined by the Scrum team during Sprint Planning.

ID	User Story / Task	Priority	Story Points	Status
US-01	User Registration	High	5	To Do
US-02	Product Search	High	5	To Do
US-03	Add Product to Cart	High	8	To Do
US-04	Place an Order	High	8	To Do
US-05	Track Order	Medium	5	To Do
T-01	Design Registration Page	High	2	To Do
T-02	Implement Registration Functionality	High	3	To Do
T-03	Develop Product Search	High	3	To Do
T-04	Develop Shopping Cart	High	5	To Do
T-05	Implement Checkout & Payment	High	5	To Do
T-06	Implement Order Tracking	Medium	3	To Do
T-07	Write Test Cases	High	5	To Do
T-08	Perform Functional Testing	High	5	To Do
T-09	Fix Identified Bugs	High	3	To Do

Sprint Backlog Structure

Sprint Duration: 2 weeks

Sprint Goal: Allow customers to register, search products, add items to the cart, place orders, and track their orders.

Workflow:

To Do → In Progress → Testing → Done

AI-Generated Output Review

Issues identified in the AI-generated Sprint Backlog:

1. Some items were written as both **user stories and development tasks**, which made the backlog inconsistent.
2. The backlog did not clearly identify the **acceptance criteria** for each user story.
3. Some tasks were too broad, such as *“Develop Shopping Cart.”*
4. The priority and story points were assigned without explaining the reasoning.
5. Testing activities could be broken down more clearly.

Manually Improved Sprint Backlog

ID	Backlog Item	Type	Priority	Story Points	Status
US-01	Register an account	User Story	High	5	To Do
US-02	Search for products	User Story	High	5	To Do
US-03	Add products to cart	User Story	High	8	To Do
US-04	Place an order	User Story	High	8	To Do
US-05	Track an order	User Story	Medium	5	To Do
T-01	Create registration form	Task	High	2	To Do
T-02	Implement product search and filtering	Task	High	3	To Do
T-03	Implement cart quantity and removal functions	Task	High	3	To Do
T-04	Implement checkout and order confirmation	Task	High	5	To Do
T-05	Create order tracking functionality	Task	Medium	3	To Do
T-06	Prepare and execute test cases	QA Task	High	5	To Do
T-07	Fix and retest identified defects	QA Task	High	3	To Do

1. What the AI did well

- **Provided a clear structure:** The AI organized the Sprint Backlog into user stories, tasks, priorities, story points, and status.
- **Used the correct Agile format:** The user stories followed the standard “As a user, I want..., so that...” format.
- **Covered important e-commerce features:** It included registration, product search, shopping cart, ordering, and order tracking.
- **Saved time:** AI quickly generated a starting point that could be reviewed and refined.
- **Suggested testing activities:** It included test-case preparation, functional testing, and bug fixing, which are important for an SQA perspective.

2. What Correction did I make

I reviewed the AI-generated Sprint Backlog and identified inconsistencies in its structure and level of detail. I separated user stories from development and QA tasks, made the tasks more specific, adjusted priorities and story points based on the expected complexity, and included testing and defect-fixing activities. The revised backlog is therefore more realistic and suitable for Sprint planning.

3. Why human review is still necessary

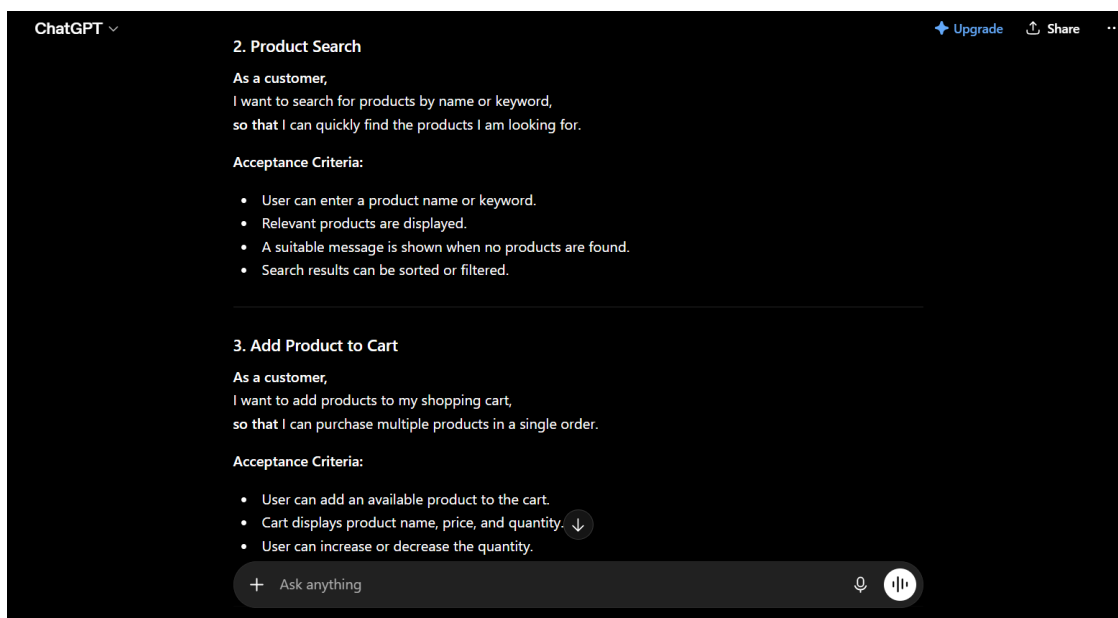
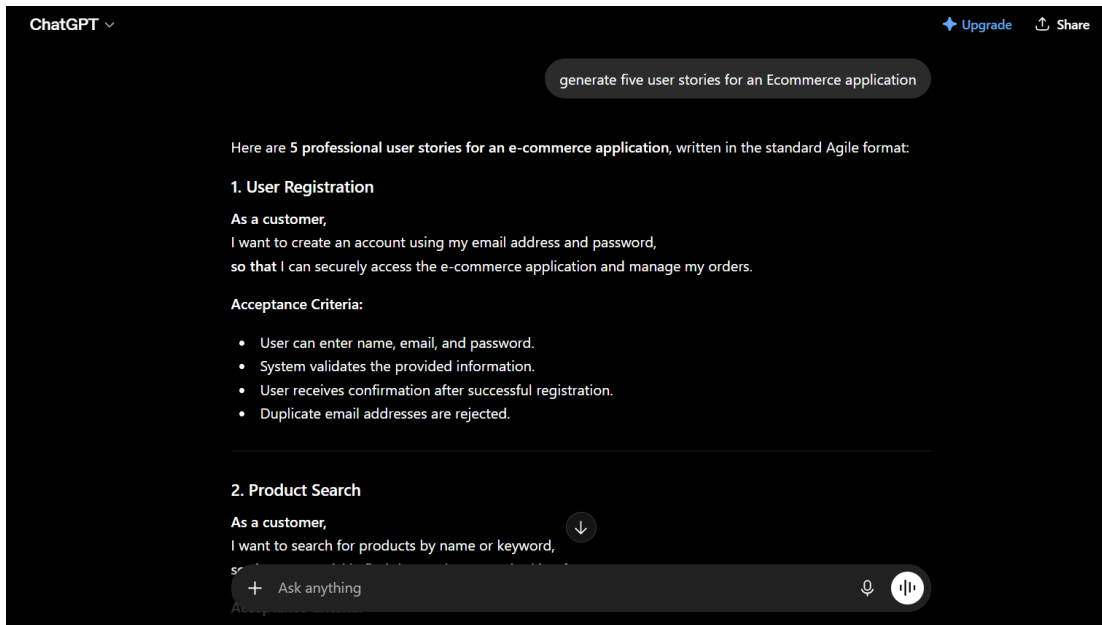
AI-generated content may be **generic or make assumptions** that do not match the actual project. A human need to check whether the backlog reflects the application's real requirements.

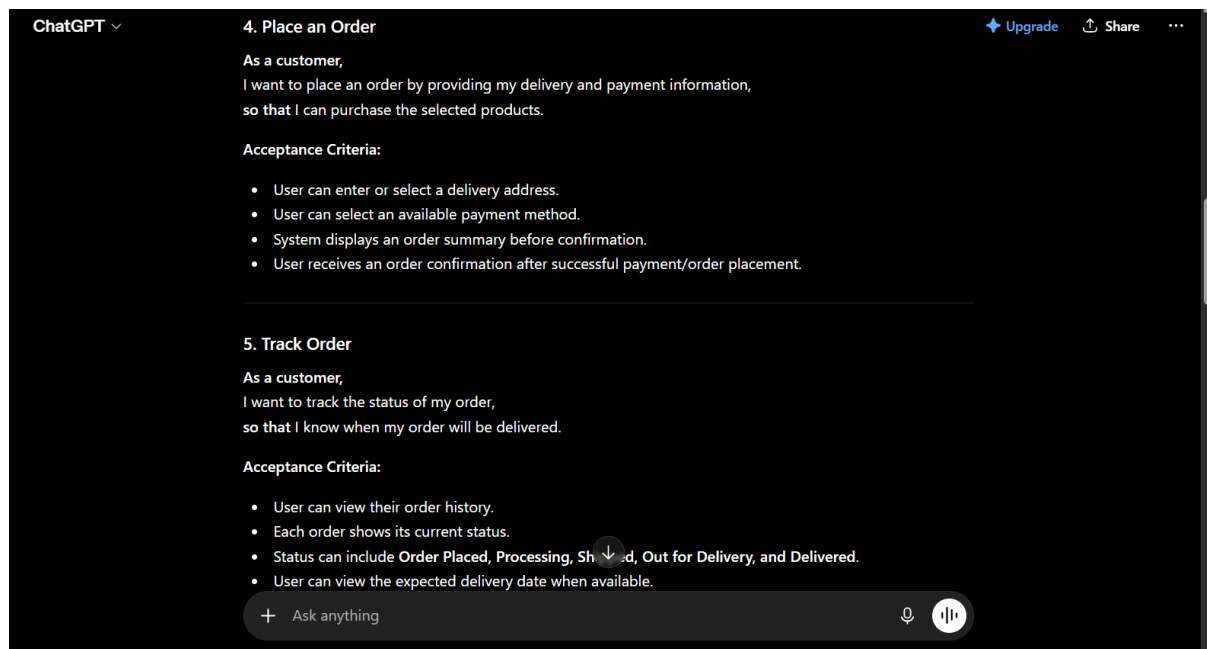
For example:

- The **priority** of a feature should be decided based on actual business requirements.
- **Story points** require judgment about the team's technical complexity and effort.
- Some tasks generated by AI may be **too broad or overlapping**.
- Acceptance criteria need to be checked to make sure they are **specific, testable, and realistic**.
- A human can identify **missing requirements**, such as payment failure handling, stock availability, or security requirements.
- The final backlog should reflect the **actual team's capacity and Sprint goal**, rather than blindly accepting AI suggestions.

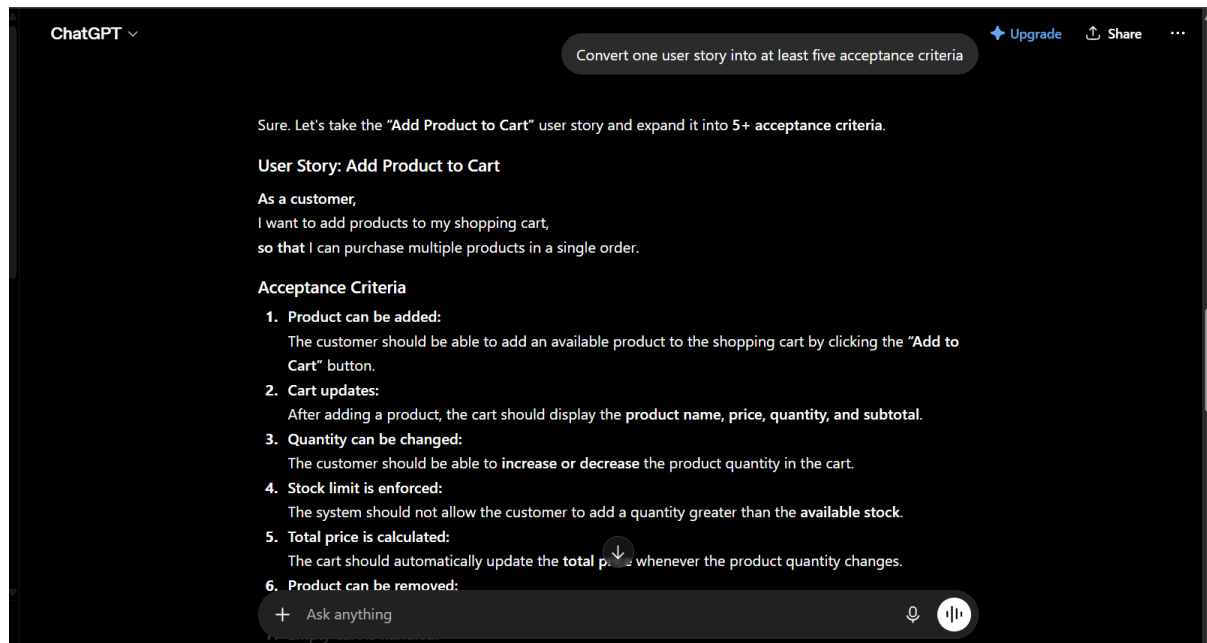
4. Screenshots of AI Interactions

Prompt 1: “Generate five user stories for an E-commerce application





Prompt 2: “Convert the checkout and payment user story into at least five clear and testable acceptance criteria.



Prompt 3: "Create a Sprint Backlog"

ChatGPT

Create a Sprint backlog

UpgradeShare

Absolutely. Based on the e-commerce user stories we created, here is a professional **Sprint Backlog** you can use for an SQA/Agile assignment.

Sprint Backlog – E-Commerce Application

Sprint Goal: Develop the basic customer shopping and ordering functionality.

ID	User Story / Task	Priority	Story Points	Status
US-01	User Registration	High	5	To Do
US-02	Product Search	High	5	To Do
US-03	Add Product to Cart	High	8	To Do
US-04	Place an Order	High	8	To Do
US-05	Track Order	Medium	5	To Do
T-01	Design Registration Page	High	2	To Do
T-02	Implement Registration Functionality	High	3	To Do

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Ask anything

🔊

ChatGPT

UpgradeShare

T-02	Implement Registration Functionality	High	3	To Do
T-03	Develop Product Search	High	3	To Do
T-04	Develop Shopping Cart	High	5	To Do
T-05	Implement Checkout & Payment	High	5	To Do
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Sprint Backlog Structure

Sprint Duration: 2 weeks

Sprint Goal: Allow customers to register, search products, add items to the cart, place orders, and track their orders.

Workflow:

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Ask anything

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